

Multi Stage Pattern Documentation

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Multi Stage Pattern

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Overview

The Multi Stage Pattern demonstrates the most sophisticated deployment capability in the Release Engine framework. It orchestrates complex, multi-stage deployments with dependencies between stages, allowing for advanced scenarios like prerequisite infrastructure, followed by multiple dependent deployments that can run in parallel or sequence.

Architecture

This pattern showcases a complex deployment topology with one prerequisite stage and three dependent stages that all depend on the prerequisite stage.

```
graph TB
    subgraph SUB ["Azure Subscription"]
        subgraph PREREQ ["Prerequisite Stage"]
            PRG["Resource Group"]
        end
    end

    subgraph DEP ["Dependent Stages (Parallel)"]
        subgraph S1 ["Dependent Stg 1"]
            RG1["Resource Group"]
            SA1["Storage Account"]
        end

        subgraph S2 ["Dependent Stg 2"]
            RG2["Resource Group"]
            SA2["Storage Account"]
        end

        subgraph S3 ["Dependent Stg 3"]
            RG3["Resource Group"]
            SA3["Storage Account"]
        end
    end

    PREREQ --> S1
    PREREQ --> S2
    PREREQ --> S3

    style SUB fill:#f3e5f5
    style PREREQ fill:#e1bee7
    style S1 fill:#c8e6c9
    style S2 fill:#c8e6c9
    style S3 fill:#c8e6c9
    style PRG fill:#ce93d8
```

```
style RG1 fill:#a5d6a7
style RG2 fill:#a5d6a7
style RG3 fill:#a5d6a7
style SA1 fill:#81c784
style SA2 fill:#81c784
style SA3 fill:#81c784
```

Pipeline Visualization

The following image shows the actual Azure DevOps pipeline execution with the multi-stage deployment pattern:



Pipeline stages showing Build phases, followed by Development environment (Prerequisite + 3 Dependent stages), Test environment (Prerequisite + 3 Dependent stages), and Production environment (Prerequisite + 3 Dependent stages)

Prerequisites

- **Azure Permissions:** Owner or Contributor access at subscription level
- **Service Principal:** Configured with subscription-level permissions for resource group and storage account creation
- **Multiple Parameter Files:** Separate parameter files for prerequisite and each dependent stage
- **Complex Configuration:** Understanding of stage dependencies and orchestration

Resources Created

Prerequisite Stage

- **Azure Resource Group** (AVM: avm/res/resources/resource-group:0.4.2)
 - Foundation resource group for dependent resources
 - Deployed once per environment

Dependent Stages (1, 2, 3)

Each dependent stage creates:

- **Azure Resource Group** (AVM: avm/res/resources/resource-group:0.4.2)
- **Azure Storage Account** (AVM: avm/res/storage/storage-account:0.27.1)
 - Deployed in parallel after prerequisite completion
 - Each with unique configurations via separate parameter files

Parameters

Prerequisite Stage Parameters

Parameter	Type	Description	Default	Required
tags	object	Tags for the resource group	{}	No
			example-	

resourceGroupName	string	Name of the prerequisite resource group	storage-rg	No
-------------------	--------	---	------------	----

Dependent Stage Parameters

Parameter	Type	Description	Default	Required
resourceLocation	string	Azure region for deployment	None	Yes
tags	object	Tags for resources	{}	No
resourceGroupName	string	Name of the resource group	example-storage-rg	No
storageAccountName	string	Name of the storage account	None	Yes

Parameter Constraints

- **resourceLocation:** Must be one of: westeurope, uksouth
- **storageAccountName:** Must be globally unique across all Azure storage accounts
- **resourceGroupName:** Must follow Azure naming conventions

Outputs

Prerequisite Stage

Output	Type	Description
resourceGroupId	string	Resource ID of the prerequisite resource group

Dependent Stages

Output	Type	Description
resourceGroupId	string	Resource ID of the stage-specific resource group
storageAccountId	string	Resource ID of the created storage account

Deployment Scope

- **Target Scope:** Subscription (all stages)
- **Service Connection Required:** Yes (subscription-level permissions)
- **Multi-Stage:** Yes (prerequisite + 3 dependent stages)
- **Parallel Execution:** Dependent stages run in parallel after prerequisite

Pipeline Configuration

Complete Stage Configuration

```

    stages:
      - template: /pipelines/01-
orchestrators/pattern.orchestrator.yml@release-engine-core
    parameters:
      patternSettings:
        name: multi_stage_pattern
        configurationFilePath: ${
parameters.deploymentSettings.configurationFilePath }}
        environments: ${
parameters.deploymentSettings.environments
}}

      patternArtifactsPath: /patterns/multi_stage_pattern
      stages:
        # Prerequisite Stage - Runs First
        - infrastructure:
            iac:
              name: prerequisite_stage
              displayName: Prerequisite Stage
              deploymentScope: Subscription
              serviceConnection: $(serviceConnection)
              iacMainFileName:
multi_stage_pattern.prerequisite.bicep
              iacParameterFileName:
/parameters/multi_stage_pattern.prerequisite.parameters.json

            # Dependent Stage 1 - Depends on Prerequisite
            - infrastructure:
                iac:
                  name: dependent_stage1
                  displayName: Dependent Stg 1
                  deploymentScope: Subscription
                  serviceConnection: $(serviceConnection)
                  iacMainFileName: multi_stage_pattern.dependent.bicep
                  iacParameterFileName:
/parameters/multi_stage_pattern.dependent1.parameters.json
                  dependsOn: prerequisite_stage
                  lastInStage: true

            # Dependent Stage 2 - Depends on Prerequisite (Parallel
with Stage 1 & 3)
            - infrastructure:
                iac:
                  name: dependent_stage2
                  displayName: Dependent Stg 2
                  deploymentScope: Subscription
                  serviceConnection: $(serviceConnection)
                  iacMainFileName: multi_stage_pattern.dependent.bicep
                  iacParameterFileName:
/parameters/multi_stage_pattern.dependent2.parameters.json
                  dependsOn: prerequisite_stage
                  lastInStage: true

            # Dependent Stage 3 - Depends on Prerequisite (Parallel
with Stage 1 & 2)
            - infrastructure:
                iac:
                  name: dependent_stage3
                  displayName: Dependent Stg 3
                  deploymentScope: Subscription
                  serviceConnection: $(serviceConnection)
                  iacMainFileName: multi_stage_pattern.dependent.bicep
                  iacParameterFileName:
/parameters/multi_stage_pattern.dependent3.parameters.json
                  dependsOn: prerequisite_stage
                  lastInStage: true

```

Key Configuration Points

- **Dependency Management:** All dependent stages depend on `prerequisite_stage`
- **Parallel Execution:** Stages 1, 2, and 3 run in parallel after prerequisite completion
- **Multiple Bicep Files:** Separate templates for prerequisite and dependent resources
- **Parameter File Mapping:** Each stage uses its own parameter file for customization

Configuration Repository Integration

Parameter Files Required

The pattern requires four separate parameter files:

1. `multi_stage_pattern.prerequisite.parameters.json`
2. `multi_stage_pattern.dependent1.parameters.json`
3. `multi_stage_pattern.dependent2.parameters.json`
4. `multi_stage_pattern.dependent3.parameters.json`

Prerequisite Stage Parameter File

```
{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentParameters.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "resourceGroupName": {
      "value": "rg-#{applicationName}-prerequisite-#{environmentAbbreviation}#"
    },
    "tags": {
      "value": {
        "Environment": " #{environmentAbbreviation}#",
        "Stage": "Prerequisite",
        "Application": " #{applicationName}#",
        "Owner": " #{primary_owner}#"
      }
    }
  }
}
```

Dependent Stage Parameter Files

```
{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentParameters.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "resourceLocation": {
      "value": " #{resourceLocation}#"
    },
    "resourceGroupName": {
      "value": "rg-#{applicationName}-stage1-#{environmentAbbreviation}#"
    },
    "storageAccountName": {
      "value": "st#{applicationAbbreviation}stgl1#{environmentAbbreviation}##{uniqueString}#"
    },
    "tags": {
```

```

      "value": {
        "Environment": "#{environmentAbbreviation}#",
        "Stage": "Dependent-1",
        "Application": "#{applicationName}#",
        "Owner": "#{primary_owner}#"
      }
    }
  }
}

```

Metadata Configuration

```

# In _config/metadata.yml
variables:
  workload: "multi_stage_pattern"      # Must match pattern name
  applicationName: "myComplexApp"      # Used across all stages
  applicationAbbreviation: "mca"        # For unique naming
  primary_owner: "devops@company.com"  # Consistent tagging

```

Environment Variables

Development Environment

```

# vars-development.yml
variables:
  environmentAbbreviation: dev
  resourceLocation: westeurope

# Unique string for storage account naming
uniqueString: "dev001"

# Development-appropriate configurations
enableAdvancedFeatures: false
storageAccountSku: Standard_LRS

```

Production Environment

```

# vars-production.yml
variables:
  environmentAbbreviation: prd
  resourceLocation: westeurope

# Unique string for storage account naming
uniqueString: "prd001"

# Production-grade configurations
enableAdvancedFeatures: true
storageAccountSku: Standard_GRS

```

Usage Examples

Complete Multi-Environment Deployment

The pipeline automatically promotes through environments:

1. **Development:** Prerequisite → 3 Parallel Dependent Stages
2. **Test:** Prerequisite → 3 Parallel Dependent Stages
3. **Production:** Prerequisite → 3 Parallel Dependent Stages

Customizing Stage Behavior

Each dependent stage can be customized via its parameter file:

- **Stage 1:** Primary application storage
- **Stage 2:** Backup and archival storage
- **Stage 3:** Analytics and reporting storage

Advanced Scenarios

Sequential Dependencies

To make dependent stages run sequentially instead of parallel:

```
# Stage 2 depends on Stage 1
- infrastructure:
  iac:
    name: dependent_stage2
    dependsOn: dependent_stage1 # Sequential dependency

# Stage 3 depends on Stage 2
- infrastructure:
  iac:
    name: dependent_stage3
    dependsOn: dependent_stage2 # Chain dependencies
```

Mixed Dependencies

Complex dependency chains:

```
# Stage 2 depends on both prerequisite AND stage 1
- infrastructure:
  iac:
    name: dependent_stage2
    dependsOn: [prerequisite_stage, dependent_stage1]
```

Conditional Stages

Using conditional deployment based on environment:

```
# Only deploy stage 3 in production
- infrastructure:
  iac:
    name: dependent_stage3
    condition: eq(variables['environmentAbbreviation'], 'prd')
    dependsOn: prerequisite_stage
```

Security Considerations

- **Subscription-Level Permissions:** Requires elevated permissions for resource group creation
- **Resource Isolation:** Each stage creates its own resource group for isolation
- **Consistent Tagging:** All resources tagged with stage information for tracking
- **Service Principal Security:** Single service principal used across all stages (consider stage-specific principals for enhanced security)

Performance Optimization

Parallel Execution Benefits

- **Reduced Deployment Time:** Dependent stages run simultaneously
- **Resource Efficiency:** Maximum utilization of Azure deployment capabilities
- **Faster Feedback:** Earlier detection of deployment issues

Optimization Strategies

- **Template Optimization:** Minimize dependencies within Bicep templates
- **Parameter Validation:** Front-load parameter validation to fail fast
- **Resource Grouping:** Group related resources within stages for efficiency

Cost Optimization

Multi-Stage Cost Management

- **Stage-Specific Tagging:** Enable cost allocation by deployment stage
- **Environment Scaling:** Different configurations per environment
- **Resource Lifecycle:** Coordinate lifecycle management across stages

Cost Allocation Strategy

```
# Tag strategy for cost tracking
tags:
  Environment: "${environmentAbbreviation}#"
  Stage: "dependent-1" # Track costs by stage
  Application: "${applicationName}#"
  CostCenter: "${cost_center}#"
  DeploymentGroup: "multi-stage-#{Build.BuildNumber}#"

```

Monitoring and Diagnostics

Multi-Stage Monitoring Strategy

- **Stage-Level Monitoring:** Each stage independently monitored
- **Cross-Stage Dependencies:** Monitor dependency health between stages
- **Deployment Tracking:** Track deployment progress across all stages
- **Rollback Coordination:** Coordinate rollbacks across dependent stages

Diagnostic Configuration

```
# Enable diagnostics per stage
variables:
  enableDiagnostics: true
  logAnalyticsWorkspace: "law-#{applicationName}#{environmentAbbreviation}#"

# Stage-specific diagnostic settings
stage1DiagnosticsEnabled: "${enableDiagnostics}#"
stage2DiagnosticsEnabled: "${enableDiagnostics}#"
stage3DiagnosticsEnabled: "${enableDiagnostics}#"

```

Troubleshooting

Common Multi-Stage Issues

Dependency Resolution Failures

Problem: Dependent stages fail because prerequisite stage outputs not available

Solution: Ensure prerequisite stage completes successfully and outputs are properly defined

Parallel Execution Conflicts

Problem: Resource naming conflicts between parallel stages

Solution: Use unique naming patterns in each stage's parameter file

Service Principal Permissions

Problem: Permissions insufficient for subscription-level operations

Solution: Ensure service principal has Contributor role at subscription level

Parameter File Validation

Problem: One of the four parameter files has validation errors

Solution: Validate each parameter file independently and ensure consistent token usage

Debugging Strategies

Stage Isolation Testing

```
# Test individual stages by commenting out others
stages:
  - infrastructure:
      iac:
        name: prerequisite_stage
        # ... prerequisite configuration

# Comment out dependent stages for isolation testing
# - infrastructure: ...
```

Dependency Visualization

Use the pipeline visualization to understand stage dependencies:

- Prerequisite stage shown as independent
- Dependent stages shown with arrows pointing from prerequisite
- Parallel execution clearly visible

Validation Commands

```
# Validate prerequisite template
az bicep build --file multi_stage_pattern.prerequisite.bicep

# Validate dependent template
az bicep build --file multi_stage_pattern.dependent.bicep

# Test prerequisite deployment
az deployment sub what-if \
  --location westeurope \
```

```

--template-file multi_stage_pattern.prerequisite.bicep \
--parameters @multi_stage_pattern.prerequisite.parameters.json

# Test dependent deployment (requires existing resource group)
az deployment sub what-if \
--location westeurope \
--template-file multi_stage_pattern.dependent.bicep \
--parameters @multi_stage_pattern.dependent1.parameters.json

```

Pattern Evolution

Graduating from Simpler Patterns

This pattern represents the evolution from:

- **Single Resource Pattern:** When you need multiple related deployments
- **Subscription Scope Pattern:** When you need complex dependencies
- **Simple Multi-Resource:** When you need orchestrated, multi-stage deployments

Advanced Variations

Consider these enhancements:

- **Cross-Region Deployment:** Deploy stages across multiple Azure regions
- **Multi-Subscription:** Deploy stages across different subscriptions
- **Conditional Stages:** Environment-specific stage activation
- **Dynamic Stage Generation:** Generate stages based on configuration

Use Cases

Complex Application Platforms

- **Multi-Tier Applications:** Separate stages for database, API, frontend
- **Microservices Platforms:** Individual stages for each service
- **Data Platforms:** Separate stages for ingestion, processing, analytics

Infrastructure Foundations

- **Network Foundation:** Hub network followed by spoke deployments
- **Security Foundation:** Security services followed by application resources
- **Monitoring Foundation:** Observability stack followed by application monitoring

Migration Scenarios

- **Lift and Shift:** Infrastructure first, then application components
- **Modernization:** Legacy resources followed by cloud-native components
- **Hybrid Deployments:** On-premises integration followed by cloud resources

Integration Patterns

With Other Release Engine Patterns

As Foundation for Applications

```
# Multi-stage as prerequisite for application deployment
stages:
  # Multi-stage infrastructure deployment
  - template: /patterns/multi_stage_pattern/workload.yml@workload

  # Application deployment using created infrastructure
  - template: /patterns/application_pattern/workload.yml@workload
  parameters:
    dependsOn: multi_stage_pattern_completion
```

Nested Multi-Stage Patterns

```
# Multiple multi-stage patterns in sequence
stages:
  # Foundation multi-stage (networking, security)
  - template: /patterns/foundation_multi_stage/workload.yml@workload

  # Application multi-stage (databases, APIs, frontend)
  - template:
    /patterns/application_multi_stage/workload.yml@workload
  parameters:
    dependsOn: foundation_completion
```

Maintenance and Updates

Template Maintenance

- **AVM Updates:** Regularly update both prerequisite and dependent AVM modules
- **Parameter Evolution:** Add new parameters as requirements evolve
- **Output Enhancement:** Extend outputs as downstream dependencies require
- **Security Patches:** Apply security updates consistently across all stages

Pipeline Maintenance

- **Dependency Review:** Regularly review and optimize stage dependencies
- **Performance Monitoring:** Monitor deployment times and optimize bottlenecks
- **Error Handling:** Enhance error handling and recovery mechanisms
- **Documentation Updates:** Keep stage documentation current with changes

Configuration Maintenance

- **Parameter File Sync:** Ensure consistency across all four parameter files
- **Token Standardization:** Maintain consistent token usage across stages
- **Environment Parity:** Ensure environment configurations remain aligned

- **Testing Strategy:** Maintain comprehensive testing for all stage combinations

Related Patterns

- **Single Resource Pattern:** Foundation for understanding basic deployments
- **Subscription Scope Pattern:** Foundation for understanding subscription-level operations
- **Application Patterns:** Consumers of multi-stage infrastructure deployments
- **Platform Patterns:** Advanced orchestration using multi-stage as building blocks

This pattern demonstrates the most advanced deployment orchestration capabilities in the Release Engine framework, enabling complex, enterprise-grade infrastructure deployments with proper dependency management and parallel execution optimization.